

Math 242, S13
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	25	
2	25	
3	25	
4	25	
Total	100	

Good Luck !

Problem 1. (25 points) Given the following initial value problem (Logistic equation):

$$\frac{dx}{dt} = 15x - 3x^2, \quad x(0) = 10$$

(a) Find all critical points of the given differential equation and then determine whether it is stable or unstable for each of them.

Solution: $15x - 3x^2 = 0$
 $x(15 - 3x) = 0 \Rightarrow 3x(5 - x) = 0$
 $\Rightarrow x = 0, x = 5 \rightarrow \text{critical points}$

$$\begin{array}{ccc} x' < 0 & x' > 0 & x' > 0 \\ \leftarrow & \rightarrow & \rightarrow \\ \leftarrow 0 & & 5 \rightarrow \\ \text{unstable} & & \text{stable} \end{array}$$

(b) Solve the initial value problem for $x(t)$.

Solution: $\frac{dx}{dt} = 3x(5 - x)$
 $\int \frac{1}{x(5-x)} dx = \int 3 dt$
 $\frac{1}{5} \int \left(\frac{1}{x} + \frac{1}{5-x} \right) dx = \int 3 dt$
 $\ln|x| - \ln|5-x| = 15t + C$
 $\ln \left| \frac{x}{5-x} \right| = 15t + C$
 $\frac{x}{5-x} = e^{15t+C}$
 $\Rightarrow \frac{x}{5-x} = A e^{15t}$
 $x(0) = 10 \Rightarrow A = \frac{10}{5-10} = -2$
 $\Rightarrow \frac{x}{5-x} = -2e^{15t}$
 $x = -10e^{15t} + 2e^{15t}x$
 $\Rightarrow x(t) = \frac{10e^{15t}}{2e^{15t} - 1} = \frac{10}{2 - e^{15t}}$

Problem 2. (25 points) Given an initial value problem

$$\frac{dy}{dx} = y + x + 1, \quad y(0) = -1.$$

Apply **Euler's method** to approximate the solution of the initial value problem on the interval $[0, 0.3]$ with step size $h = 0.1$.

Solution: $f(x, y) = y + x + 1$, $x_0 = 0$, $y_0 = -1$, $h = 0.1$.

$$x_1 = x_0 + h = 0.1$$

$$\begin{aligned} y_1 &= y_0 + f(x_0, y_0) \cdot h = y_0 + (y_0 + x_0 + 1) \cdot h \\ &= -1 + (-1 + 0 + 1) \cdot (0.1) \\ &= -1 \end{aligned}$$

$$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$$

$$\begin{aligned} y_2 &= y_1 + f(x_1, y_1) \cdot h = y_1 + (y_1 + x_1 + 1) \cdot h \\ &= -1 + (-1 + 0.1 + 1) \cdot (0.1) \\ &= -1 + 0.01 = -0.99 \end{aligned}$$

$$x_3 = x_2 + h = 0.3$$

$$\begin{aligned} y_3 &= y_2 + f(x_2, y_2) \cdot h = y_2 + (y_2 + x_2 + 1) \cdot h \\ &= -0.99 + (-0.99 + 0.2 + 1) \cdot (0.1) \\ &= -0.99 + 0.021 \\ &= -0.969 \end{aligned}$$

Problem 3. (25 points) Given the following homogeneous second-order differential equation:

$$y'' + 6y' + 13y = 0.$$

(a) Find the general solution of the given differential equation.

Solution: Characteristic equation.

$$r^2 + 6r + 13 = 0$$

$$r = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 1 \cdot 13}}{2} = \frac{-6 \pm \sqrt{-16}}{2}$$

$$= -3 \pm 2i$$

$$\Rightarrow y(x) = e^{-3x}(C_1 \cos 2x + C_2 \sin 2x).$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 2, \quad y'(0) = 0.$$

$$\begin{aligned} \text{Solution: } y'(x) &= (-3)e^{-3x}(C_1 \cos 2x + C_2 \sin 2x) \\ &\quad + e^{-3x}(-2C_1 \sin 2x + 2C_2 \cos 2x) \\ &= e^{-3x} [(-3 \cos 2x - 2 \sin 2x)C_1 + (-3 \sin 2x + 2 \cos 2x)C_2] \end{aligned}$$

$$\begin{aligned} y(0) = 2 &\Rightarrow y(0) = C_1 = 2 \\ y'(0) = 0 &\Rightarrow y'(0) = -3C_1 + 2C_2 = 0 \end{aligned} \quad \left\{ \Rightarrow \begin{aligned} C_1 &= 2 \\ C_2 &= 3 \end{aligned} \right.$$

Thus, we have

$$y(x) = e^{-3x}(2 \cos 2x + 3 \sin 2x)$$

Problem 4. (25 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} - 4y'' + 4y' = 3x - 1.$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

Solution: Its characteristic equation

$$r^3 - 4r^2 + 4r = 0$$

$$r(r^2 - 4r + 4) = 0$$

$$r(r-2)^2 = 0$$

$$\Rightarrow r_1 = 0, r_2 = r_3 = 2$$

$$\begin{aligned}\Rightarrow y_c(x) &= C_1 + (C_2 + C_3 x) e^{2x} \\ &= C_1 + C_2 e^{2x} + C_3 x e^{2x}\end{aligned}$$

(b) Then find a particular solution y_p of the nonhomogeneous equation.

Solution: We have $f(x) = 3x - 1$, thus we set

$$y_p(x) = x^s (Ax + B)$$

$\hookrightarrow s=1$ to avoid duplication of terms in $y_c(x)$.

$$\Rightarrow y_p(x) = x(Ax + B) = Ax^2 + Bx$$

$$\Rightarrow y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

$$y_p'''(x) = 0$$

$$\begin{aligned}y_p^{(3)}(x) - 4y_p''(x) + 4y_p'(x) &= 0 - 4(2A) + 4(2Ax + B) \\ &= \underline{\underline{-8A + 8Ax + 4B}} \equiv \underline{\underline{3x - 1}}\end{aligned}$$

$$\Rightarrow \begin{cases} 8A = 3 \\ -8A + 4B = -1 \end{cases} \Rightarrow \begin{cases} A = \frac{3}{8} \\ B = \frac{1}{2} \end{cases}$$

$$\text{thus } y_p(x) = \frac{3}{8}x^2 + \frac{1}{2}x$$